



NAFIS AHMED

Electrical & Electronic Engineer | PLC & Automation | Power Systems | Solar PV

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PROFESSIONAL SUMMARY

Electrical & Electronic Engineer (B.Sc.) with certified hands-on training in PLC programming, HMI development, and industrial instrumentation through an ADB-funded competency-based programme. Practical experience with Wecon LX3V and Siemens S7-1200 hardware — digital I/O wiring, AI/AO modules, RTD PT-100 scaling, PID control in ladder logic, 0–10V and 4–20mA analog processing, VFD control via analog output, and Ethernet commissioning via TIA Portal. Certified in Solar PV design and Power BI. Proficient in AutoCAD Electrical, MATLAB, and Proteus. Targeting roles in Automation, Electrical Maintenance, Instrumentation, Power Systems, Solar PV, or Technical Support Engineering.

TECHNICAL SKILLS

PLC & Automation

- Wecon LX3V (HW, BD Board, DA/AD)
- Siemens S7-1200 (TIA Portal —
- HW Wiring, AI/AO, PID, Ethernet)
- Ladder Logic | PID Control (PLC)
- PLC Troubleshooting & Commissioning
- RTD Scaling | 0–10V & 4–20mA
- VFD Control via Analog Output
- Delta VFD M-Series
- Automation Studio | Connect Master

HMI & Communication

- Wecon HMI & PiStudio
- Modbus RTU | RS-422 | RS-485

Control & Field Devices

- Relay | Timer | Counter | SSR
- Temperature Controller | Contactor
- Solenoid Valve | Pneumatic Cylinder
- Limit / Float Switch | Float Sensor
- Photoelectric Sensor
- Proximity (Capacitive & Inductive)

Instrumentation & Test

- RTD PT-100 | Thermocouples | Tachometer
- Pressure & Level Transmitters
- Magnetic Float Transmitter
- Earth Tester | Megger | Clamp Meter

Solar PV

- Off/On-Grid & Hybrid System Theory
- Tilt, Irradiance & Shading Analysis
- SLD, Protection & Net Metering
- ROI & Feasibility Analysis
- Solar Atlas | NASA POWER App

Software & Data

- AutoCAD 2D Electrical | TIA Portal
- MATLAB | Proteus
- Power BI Dashboard (Certified)
- MS Office Suite

Programming

- C++ | Embedded C (Basic)

Tools

- Hand & Power Tools | Panel Wiring

HSE

- OHS | PPE | 6R Green Practices

Soft Skills

- Analytical | Detail-Oriented | Fast Learner
- Self-Motivated | Team Player | Adaptable

KEY PROJECTS

Siemens S7-1200 Analog & PID Control: Wired digital I/O and AI/AO modules; scaled RTD PT-100, processed 0–10V/4–20mA signals, implemented PID in ladder logic, controlled VFD via analog output. Uploaded via Ethernet, TIA Portal.

Phase Protection System: Designed three-phase motor protection using phase-failure relay for phase loss, reversal, and under-voltage; operated magnetic contactors for motor isolation.

Motor Control with Temperature Monitoring: Programmed Wecon LX3V PLC for motor operation with RTD PT-100 monitoring and live HMI display.

Automatic Pump Control System: Level-based pump automation using float switch and float sensor for automatic start/stop control.

Conveyor Belt Automation: Start/stop, emergency stop, object detection sensor, and HMI object counter using PLC.

Tank Level Monitoring & Control: PLC tank control with level transmitter, pump automation, high/low alarms, and HMI display.

VFD Motor Speed Control: PLC-controlled Delta VFD speed via HMI with configurable ramp-up and ramp-down.

Alarm & Fault Monitoring System: PLC fault monitoring for overload, over-temperature, sensor failure with HMI alarm acknowledgment.

Traffic Light Control System: Timed sequencing using relay logic circuit and PLC ladder logic programming.

Star-Delta & Reverse-Forward Control: Auto star-delta starter and forward-reverse circuit with counter and relay logic.

DC Motor Speed Control: Microcontroller-based PWM speed regulation programmed in Embedded C.

TRAINING & CERTIFICATION

Industrial PLC & Automation — Certificate Mar–Jun 2026

SIMT, Dhaka | ADB-Funded SICIP | Competency-Based, Fully Practical

- Wecon LX3V, Siemens S7-1200 (hardware, AI/AO, PID, Ethernet via TIA Portal), Delta VFD, HMI, Modbus; instrumentation, control devices, OHS & 6R.

Solar PV System Design — Certificate 2026

Structured Online Programme (30 Days)

- PV fundamentals, battery, PWM/MPPT, inverter sizing, off/on-grid/hybrid, SLD, net metering, ROI; Solar Atlas & NASA POWER App.

Power BI Dashboard Design — Workshop Certificate 2026

3-Hour Certified Workshop

- Data visualisation, dashboard creation, and report building in Microsoft Power BI.

WORK EXPERIENCE

Technical & Operations Intern 2018–2019 · 6 months

Oriental Power Co. — Dhaka

- Hands-on exposure to generators, distribution panels, and electrical components under senior engineers.

Power Section Intern 2021–2022 · 4 months

Fiber At Home — Bangladesh

- Supported power section using Connect Master for rack management; coordinated with field technicians and met deadlines.

Marketing Executive 2023–2024 · 1 year

Powermann Bangladesh Ltd. — Dhaka

- Managed client database; cold-called for HT/LT panels, PFI panels, transformers; visited sites and attended cross-department meetings.

EDUCATION

B.Sc. – Electrical & Electronic Engineering 2018

Independent University, Bangladesh (IUB) | CGPA: 2.61/4.00

H.S.C. – Science 2011

B.A.F. Shaheen College, Tejgaon | GPA: 4.10/5.00

S.S.C. – Science 2009

Bangladesh Navy School | GPA: 5.00/5.00

Languages: Bengali — Native | English — Professional | Hindi — Conversational (spoken only) **Interests:** Travelling | Music | Sports | Reading